U5_AT3 REVERSE ENGINEERING

OPTIROUTE: Intelligent Delivery Route Optimisation System — Reverse Engineering

1. System Decomposition & Analysis

The OptiRoute: Intelligent Delivery Route Optimisation System can be decomposed into several major components: the user interface, delivery and house dataset, road-network module, construction/road-restriction module, route optimisation engine, AI decision-making module, database, and result visualisation module. The system receives information such as delivery locations, unique door numbers, streets, road connections, vehicle restrictions, and delivery requirements. This data is stored and processed by the backend, while the optimisation engine analyses possible routes and selects an efficient route that visits the required houses while avoiding restricted or blocked roads. The system therefore follows a pipeline of input data → preprocessing → road-network construction → route optimisation → restriction checking → optimal route generation → visualisation. This decomposition makes the system easier to understand, maintain, and improve.

2. Architecture Reconstruction

The reconstructed architecture of OptiRoute follows a layered architecture consisting of the presentation layer, application/backend layer, optimisation layer, and database layer. The user interacts with the web interface to provide delivery information and road conditions. The backend receives these inputs and communicates with the database to retrieve house, street, and restriction information. The optimisation engine then represents the streets and houses as a graph, where houses or important locations act as nodes and connecting roads act as edges. Restricted or construction-affected roads are marked as unavailable or assigned a very high cost. The route-planning algorithm processes this graph and generates the best feasible delivery sequence. Finally, the calculated route is returned to the frontend and displayed visually, allowing the user to understand the recommended path and affected roads.

3. Algorithm Identification

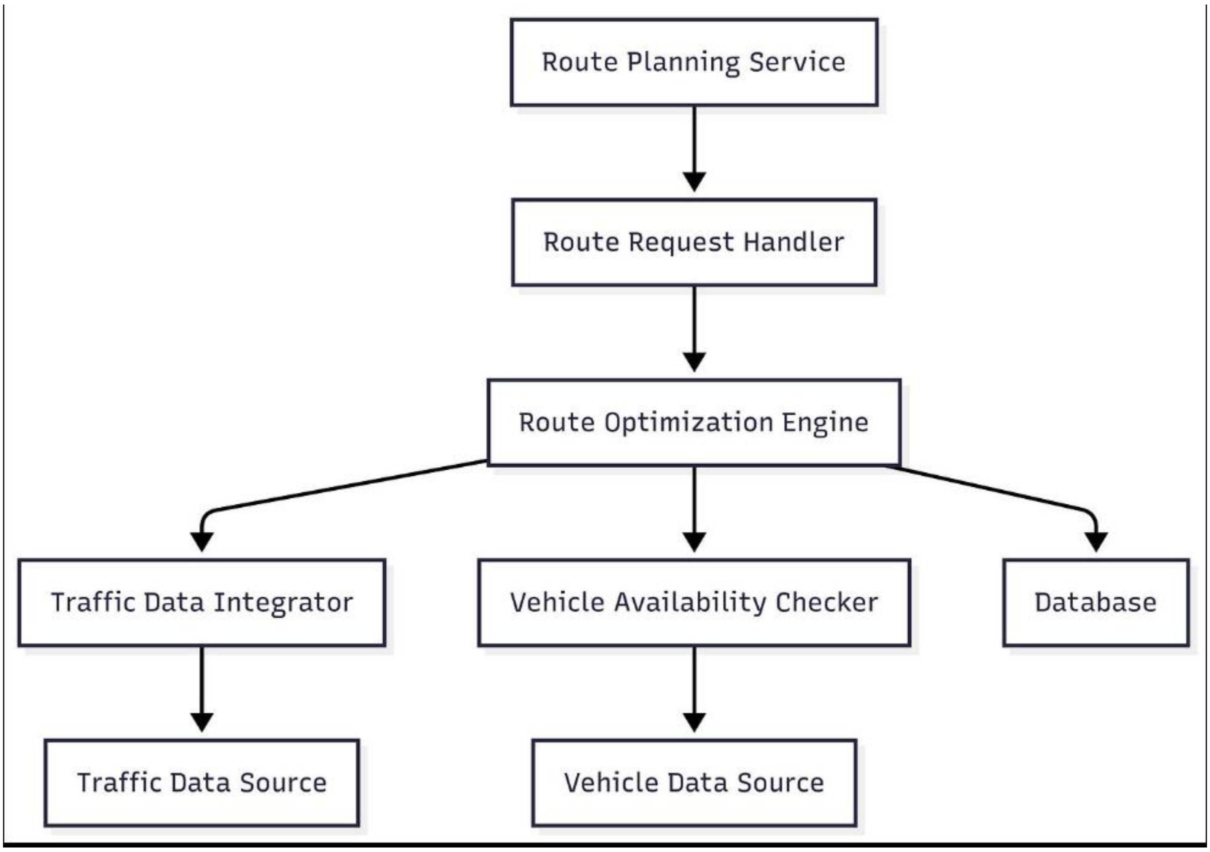
From the expected behaviour of OptiRoute, the core system can be reverse engineered as a graph-based route optimisation application. The road network can be represented using graph theory, with locations represented as vertices and streets represented as edges containing distance or travel cost. A shortest-path algorithm such as Dijkstra's algorithm or A* can be used to determine efficient paths between individual delivery locations. Since the system needs to visit multiple houses, the overall problem resembles the Travelling Salesman Problem (TSP) or a Vehicle Routing Problem (VRP). A heuristic or optimisation technique can therefore determine an efficient delivery sequence instead of checking every possible combination. Before optimisation, the system processes the dataset, identifies valid roads, detects restricted streets, and removes or penalises unavailable connections. The decision-making process then compares possible routes according to factors such as total distance, travel time, number of restricted roads, and delivery coverage before selecting the most suitable route.

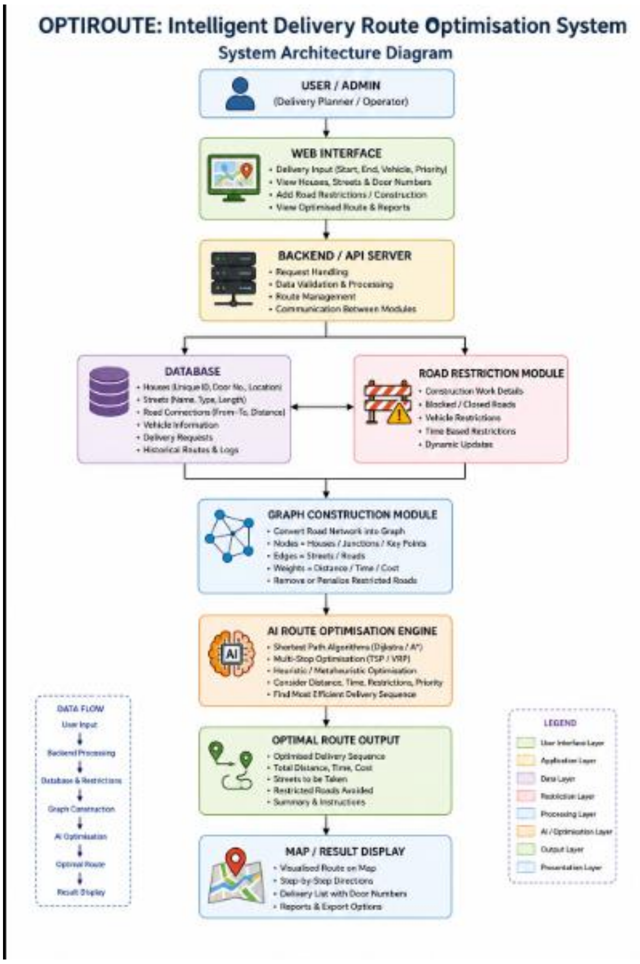
4. Data Flow & Process Mapping

The data flow in OptiRoute begins when the system loads the delivery dataset containing houses, door numbers, streets, and road connections. Additional information regarding construction work or vehicle restrictions is then associated with the corresponding streets. The preprocessing module validates the data and converts it into a structured road graph. This graph is passed to the route optimisation engine, which analyses the available connections and calculates a suitable delivery sequence. During optimisation, restricted roads are avoided, while alternative roads are considered. The resulting route information, including the sequence of houses, streets used, total distance, and restrictions encountered, is sent back to the backend and then to the frontend. The frontend displays this information on the map or route interface. Thus, the complete process can be represented as Dataset → Data Validation → Road Graph → Restriction Analysis → Route Optimisation → Optimal Route → Map/Result Display.

5. Improvement & Redesign

The reverse-engineered OptiRoute system can be improved by introducing a more scalable and intelligent architecture. The optimisation engine can be separated from the main backend so that complex route calculations do not slow down the user interface. Algorithms such as A* can improve shortest-path calculations, while advanced optimisation techniques such as Genetic Algorithms, Ant Colony Optimisation, or Machine Learning-based prediction can be used for larger delivery networks. The system can also incorporate real-time traffic, changing construction conditions, vehicle capacity, delivery priorities, and estimated travel times. Database indexing and caching can improve response speed, while an API-based modular architecture can allow future integration with mapping and GPS services. A redesigned system could therefore continuously update the road network and dynamically recalculate routes whenever a road becomes unavailable, making OptiRoute more efficient, scalable, adaptive, and suitable for real-world intelligent delivery management.


Route Planning Service
Route Request Handler
Route Optimization Engine
Traffic Data Integrator
Vehicle Availability Checker
Database
Traffic Data Source
Vehicle Data Source



OPTIROUTE: Intelligent Delivery Route Optimisation System
System Architecture Diagram
USER / ADMIN
(Delivery Planner / Operator)
WEB INTERFACE
• Delivery Input (Start, End, Vehicle, Priority)
• View Houses, Streets & Door Numbers
• Add Road Restrictions / Construction
• View Optimised Route & Reports
BACKEND / API SERVER
• Request Handling
• Data Validation & Processing
• Route Management
• Communication Between Modules
DATABASE
• Houses (Unique ID, Door No., Location)
• Streets (Name, Type, Length)
• Road Connections (From-To, Distance)
• Vehicle Information
• Delivery Requests
• Historical Routes & Logs
ROAD RESTRICTION MODULE
• Construction Work Details
• Blocked / Closed Roads
• Vehicle Restrictions
• Time Based Restrictions
• Dynamic Updates
GRAPH CONSTRUCTION MODULE
• Convert Road Network into Graph
• Nodes = Houses / Junctions / Key Points
• Edges = Streets / Roads
• Weights = Distance / Time / Cost
• Remove or Penalise Restricted Roads
AI ROUTE OPTIMISATION ENGINE
• Shortest Path Algorithms (Dijkstra / A*)
• Multi-Stop Optimisation (TSP / VRP)
• Heuristic / Metaheuristic Optimisation
• Consider Distance, Time, Restrictions, Priority
• Find Most Efficient Delivery Sequence
OPTIMAL ROUTE OUTPUT
• Optimised Delivery Sequence
• Total Distance, Time, Cost
• Streets to be Taken
• Restricted Roads Avoided
• Summary & Instructions
MAP / RESULT DISPLAY
• Visualised Route on Map
• Step-by-Step Directions
• Delivery List with Door Numbers
• Reports & Export Options
DATA FLOW
User Input
Backend Processing
Database & Restrictions
Graph Construction
AI Optimisation
Optimal Route
Result Display
LEGEND
User Interface Layer
Application Layer
Data Layer
Restriction Layer
Processing Layer
AI / Optimisation Layer
Output Layer
Presentation Layer